ESILV - Tutorial #2

Commodity Markets and Models

- This session (worth **5%**) lasts only 1h30.
- Solve exercises **on paper** and hand it in to your tutorial leader (*one per group*) before the end. (no email) – (no late submission)
- Ask your instructor for advice. Corrections will not be posted on DVL.
- In case of absence to this tutorial, you will be graded as zero. **Your instructor does not need to know the reason of your absence.** Official absence needs to be recorded directly with the scholar office (`scolarite-esilv@devinci.fr`).

Problem 1: `Stochastic model for oil`

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\{\mathcal{F}_t\}_{t\geq 0}$, where $\mathcal{F}_t = \sigma(X_s : 0 \leq s \leq t)$. The log-spot price $s_t = \log(S_t)$ follows:

$$ds_t = \mu\, dt + \sigma\, dW_t, \quad t \geq 0$$

where W_t is a standard Brownian motion adapted to $\mathcal{F}_t$, μ is the constant drift, and $\sigma > 0$ is the constant volatility.
The discrete-time approximation for daily log-returns $\Delta s_t = \log\left(\frac{S_t}{S_{t-1}}\right)$, assuming a time step $\Delta t = 1$, is:

$$\Delta s_t = \mu + u_t, \quad u_t = \sigma\varepsilon_t, \quad \varepsilon_t \sim \mathrm{IID}(0,1), \quad t = 1, \ldots, T$$

with $\varepsilon_t = \Delta W_t$ and $\varepsilon_t \perp \varepsilon_s$ for $t \neq s$.
The log-likelihood function for this model is:

$$\mathcal{L}(\mu, \sigma) = -\frac{T}{2}\log(2\pi) - T\log(\sigma) - \frac{1}{2\sigma^2}\sum_{t=1}^{T}(\Delta s_t - \mu)^2$$

1) Write the *pseudo-code* to program this log-likelihood function, either in Python or R.

2) Comment on the resulting maximum likelihood estimates for the parameters (Table 1).

Next, consider the GARCH(1,1) model to capture volatility clustering. The volatility process is:

$$\sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta\sigma_{t-1}^2$$

with constraints: $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta < 1$ for stationarity. Here, u_t represents the return innovation (residual). The log-likelihood function for the GARCH(1,1) model is:

$$\mathcal{L}(\theta) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(\sigma_t^2\right) - \frac{1}{2}\sum_{t=1}^{T}\frac{(\Delta s_t - \mu)^2}{\sigma_t^2}$$

where $\theta = \{\mu, \omega, \alpha, \beta\}$ and σ_t^2 follows the GARCH(1,1) recursion.

3) Write the *pseudo-code* for the log-likelihood of the oil price model with GARCH(1,1) volatility. *Note: You must specify the initialization of* σ_0^2.

4) Comment on the corresponding parameter estimates (Table 2). What does the persistence parameter $\alpha + \beta$ tell us? How does this model improve upon the constant volatility model?

5) Comment on the graph showing 250 out-of-sample oil price forecasts with $\pm 2\sigma$ volatility bands (Figure 1). What does this visualization tell us about forecast uncertainty and the GARCH model performance?

End of Exercise. Please hand in your exercise on paper to your instructor.

Parameter	Estimate	Std. Error	t-value	p-value
μ	0.0018	0.0005	3.60	0.0003
σ	0.026	0.001	26.00	< 0.0001
Model Diagnostics:				
Observations (T)				1000
Log-Likelihood				2712.4
AIC				-5420.8

Table 1: Parameter estimates for the constant volatility model.

Parameter	Estimate	Std. Error	t-value	p-value
μ	0.0016	0.0004	4.00	0.0001
ω	0.000012	0.000004	3.00	0.0027
α	0.086	0.015	5.73	< 0.0001
β	0.902	0.012	75.17	< 0.0001
Model Diagnostics:				
Observations (T)				1000
Log-Likelihood				2820.3
AIC				-5632.6
Persistence ($\alpha + \beta$)				0.988
Unconditional variance				0.001

Table 2: GARCH(1,1) parameter estimates.

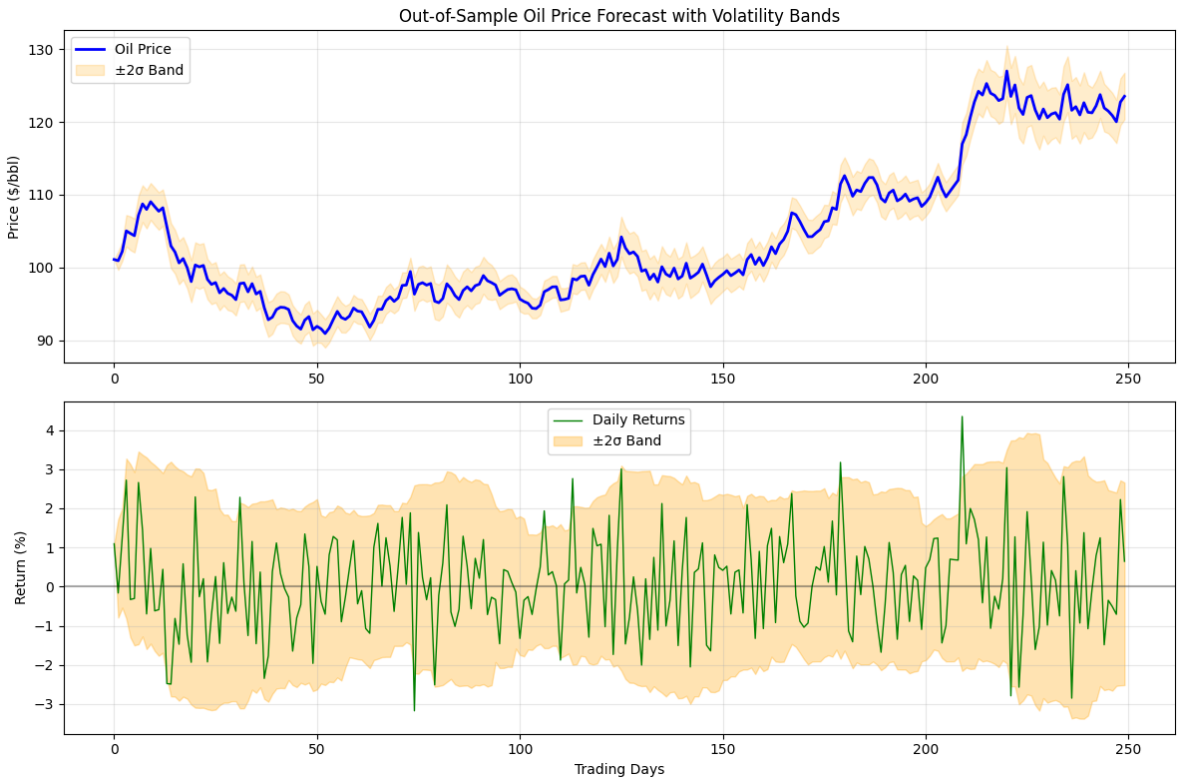

Figure 1: Out-of-sample oil price forecasts (250 trading days) with $\pm 2\sigma$ volatility bands from GARCH(1,1) model.